

## Multiple Choice (10 marks)

1. Say the pupil of your eye has a diameter of 5 mm and you have a telescope with an aperture of 50 cm. How much more light can the telescope gather than your eye?
  - (a) 10000 times more
  - (b) 100 times more
  - (c) 1000 times more
  - (d) 10 times more
2. What is true about the rotation of a solid disk?
  - (a) The velocity is the same on every point on the disk
  - (b) The part closer to the axis has a longer period
  - (c) The period of rotation is the same on every point on the disk
  - (d) The part closer to the axis has a shorter period
3. The resolution of a telescope is a measure of its:
  - (a) Ability to magnify distant objects
  - (b) Ability to measure the angular separation of objects
  - (c) Ability to measure the distance between objects
  - (d) Light-collecting efficiency
4. A comet of mass  $m$  impacts the earth (mass  $M$  radius  $R$ ) at the minimum impact speed. What is the expression for the total energy released in the impact?
  - (a)  $m \cdot v$
  - (b)  $0.5 \cdot m / (R^3)$
  - (c)  $0.5 \cdot m \cdot (2 GM/R)$
  - (d)  $0.6 \cdot G(M^2)/R$
5. Find the best approximation for the surface temperature of the Sun:
  - (a) 6000 K
  - (b) 7000 K
  - (c) 9000 K
  - (d) 13000 K

## True / False (10 marks)

*Answer True or False*

6. True or False: The atmosphere of Mars is mostly nitrogen, which is suitable for human breathing.
7. True or False: The blue color of the sky results from the preferential scattering of shorter wavelengths of light by atmospheric molecules.
8. True or False: Mars has more carbon dioxide in its atmosphere than Earth, which contributes significantly to its seasonal variations.
9. True or False: The ecliptic is the apparent path of the Sun across the sky as observed from Earth throughout the year.
10. True or False: The Cassini division is a significant gap in Saturn's rings, created by orbital resonance with the moon Mimas.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the various surface processes that shape Venus, analyzing their implications for understanding the planet's geology and atmospheric dynamics.
12. Discuss the implications of Mars' orbital period in relation to Earth years, particularly focusing on how this difference affects potential future colonization and exploration efforts.

*End of Paper*